Security Overhaul | Project Overview

Fairway Accounting Group

A practical security assessment and remediation lab for a fictional 45-employee accounting firm.

10-MINUTE TECHNICAL PRESENTATION

Focus

Network security, identity, monitoring and verification

Approach

Assess → Improve → Test → Document

Business Context & Initial Security Concerns

Business context

Accounting, payroll and tax services

Sensitive financial and client records

Growing organization with 45 employees

Initial weaknesses

Broad internal access

Excessive Payroll permissions

Weak password and lockout settings

HTTP portal and limited logging

Project objective

Reduce unnecessary access

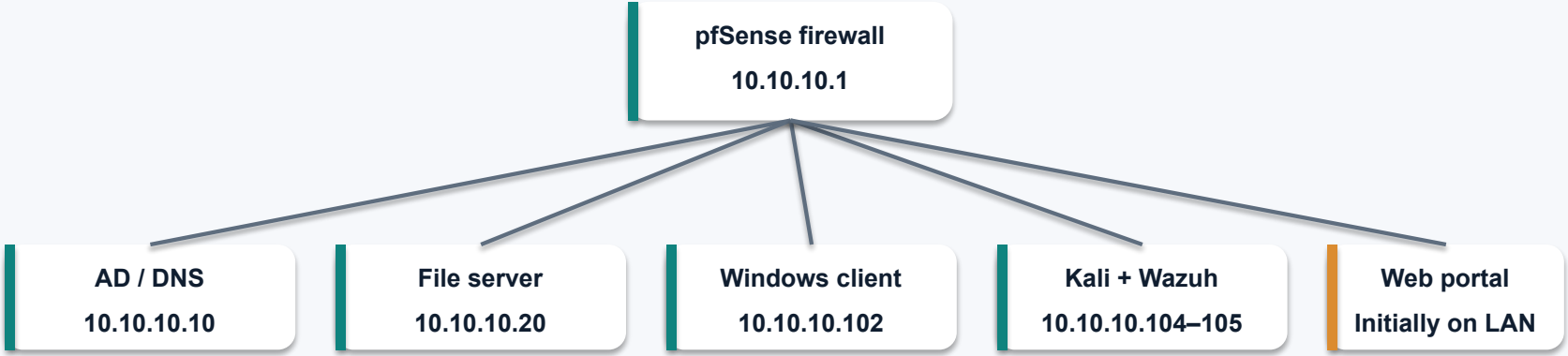
Improve authentication and transport security

Add centralized visibility

Verify changes with evidence

Scope: representative lab systems and controlled tests — not all 45 production users.

Initial Network | What Needed Attention



Main concern: important systems shared broad network access, and several controls needed hardening.

Risk Assessment | Prioritizing the Work

Higher-priority risks

Excessive Payroll file access

Unencrypted HTTP portal

Weak domain authentication settings

Limited monitoring coverage

Risk treatment

Remove incorrect group membership

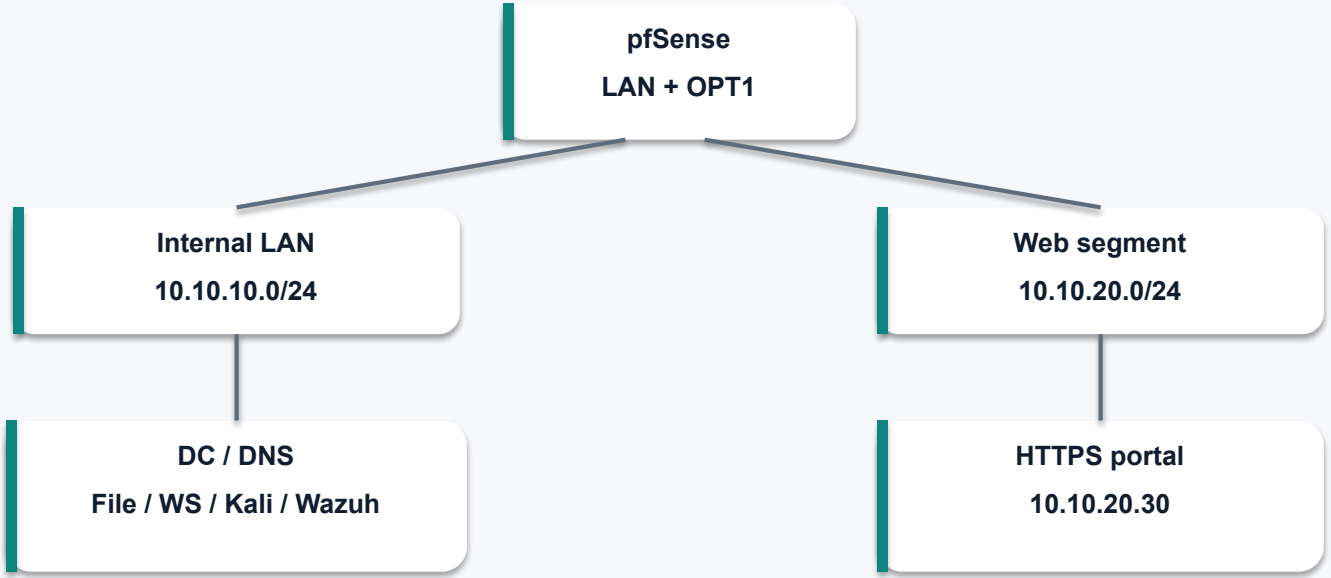
Enable HTTPS and restrict SSH

Harden password and lockout policies

Enroll a Windows endpoint in Wazuh

Eight risks assessed: 6 High and 2 Medium initially; 6 Medium and 2 Low after remediation.

Secured Network | Targeted Segmentation



HTTPS 443 ALLOWED

SSH 22: KALI ONLY

Important limitation: LAN rules remained broad; this was partial segmentation.

Security Improvements | Before → After

Payroll access	Incorrect group membership	→	**Removed; access denied**
Authentication	Minimum 7 / no lockout	→	**Minimum 12 / lockout at 5**
Client portal	HTTP open / HTTPS closed	→	**HTTPS open / HTTP filtered**
File sharing	SMB signing not required	→	**SMB signing required**

Testing & Validation | Evidence-Based Results

Verified results

PORTAL SCAN

443/tcp — Open (HTTPS)

80/tcp — Filtered (HTTP)

22/tcp — Open from Kali (SSH)

AD ACCESS

Payroll access denied after fix

FILE SHARING

SMB signing required

Evidence #44 | Final portal Nmap scan

```
┌──(securityadmin㉿fw-kali01)-[~]
└─$ sudo nmap -Pn -sV -p 22,80,443 10.10.20.30
[sudo] password for securityadmin:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-06 21:27 -0400
Nmap scan report for 10.10.20.30
Host is up (0.0011s latency).

PORT    STATE    SERVICE  VERSION
22/tcp  open     ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.19 (Ubuntu Linux; protocol 2.0)
80/tcp  filtered http
443/tcp open     ssl/http Apache httpd 2.4.58 ((Ubuntu))
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.03 seconds
```

Scan target: 10.10.20.30 • Executed from FW-KALI01

Additional connectivity and service checks were saved using a Bash script.

Selected control validation — not a comprehensive vulnerability assessment or penetration test.

Security Monitoring | Controlled Login Simulation

Observed results

TEST
3 intentional failed logins
FAIRWAY\acarter

WINDOWS EVENT
Event ID 4625

WAZUH DETECTION
3 alerts • Rule 60122
Severity level 5
One Windows endpoint enrolled

Evidence #41 | Wazuh authentication alerts

3 hits

Oct 6, 2026 @ 20:29:17.274 - Oct 6, 2026 @ 20:54:27.765

Export Formatted | Reset view | 643 available fields | Columns | Density | **2 fields sorted**

Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
Oct 6, 2026 @ 20:30:52.190	FS-WS01	Logon Failure - Unknown user or bad pa...	5	60122
Oct 6, 2026 @ 20:30:46.629	FS-WS01	Logon Failure - Unknown user or bad pa...	5	60122
Oct 6, 2026 @ 20:30:33.857	FS-WS01	Logon Failure - Unknown user or bad pa...	5	60122

Three Level 5 alerts recorded by Wazuh rule 60122.
Agent label: FS-WS01 (Windows workstation).

Controlled detection test only — not a confirmed breach or full incident response exercise.

Troubleshooting | What I Learned

DNS resolution

Symptom: workstation could not resolve an external Wazuh download hostname.

Fix: corrected DNS forwarding on the domain controller.

Network controls

Checked addressing, gateways, pfSense policy, and host firewall behavior after separating the portal.

Service health

Checked Wazuh manager, indexer, and dashboard services and addressed setup issues.

Lesson: isolate the failing layer, make one targeted change, then retest.

Recommendations & Conclusion

Next priorities

Extend VLAN segmentation

Roll out MFA

Expand Wazuh endpoint coverage

Use production-trusted TLS certificates

Test backup recovery and IR procedures

Project outcomes

Built and troubleshot a realistic lab

Corrected selected access and configuration risks

Verified changes with repeatable tests

Documented results and residual risks

Core takeaway: security improvements matter when they can be explained and verified.

Questions?